

Continuous-Time Portfolio Optimization: The Merton Problem

Abstract

In this paper we investigate the Merton portfolio problem in continuous time. We aim to determine and study how an investor should perform dynamic allocation of wealth between a risk-free asset and a risky asset in order to maximize the terminal wealth. We first derive the Hamilton-Jacobi-Bellman (HJB) equation and obtain a closed-form solution for the optimal investment strategy. We observe that the optimal proportion of wealth invested in the risky asset is constant over time and depends on the risk premium, volatility, and the investor's risk aversion. Further, we incorporate Monte Carlo simulations to validate our analytical results. To study the distribution of the terminal wealth, asset price paths are simulated. We also conduct sensitivity analysis to examine the effects of parameters like risk aversion, volatility, Investment Horizon, and Risk premium on the model. Through the unified analytical and numerical approach we provide an inclusive framework to understand the optimal investment behavior in continuous time.

1 Introduction

The financial market echoes profits, losses, growth, and uncertainty. One can make wise investment decisions by observing the market movement. Proper analysis of market maneuvers allows one to allocate their capital across various assets. This reduces the overall risk. Hence, an efficient financial model reflecting these aspects of a market plays a crucial role in maximizing results.

Portfolio optimization is one of the well-developed financial theories. The foremost ground-breaking contribution was made by Harry Markowitz [1] in the year 1952. Markowitz provided a rigorous mathematical footing for portfolio selection, which regularized the idea of investors evaluating a portfolio based on the trade-offs between expected return and risk instead of considering assets in isolation. Following this framework, William Sharpe [3] develops the "Capital Asset Pricing Model" (CAPM) to explain Markowitz's model in an actual market. CAPM essentially explains why assets have the returns they do based on their risk relative to the market. Following this Merton Miller and Franco Modigliani [4] established a theory to associate market values and asset price. Harry Markowitz, William Sharpe and Merton Miller were awarded the Nobel prize in economics in the year 1990 [7] [8]. As a result of which a requirement of a scientific financial theory came to the lime light.

Developing on the static setup Robert C. Merton in 1971 extends this theory to a dynamic system by considering the continuous-time setting. Merton modeled asset prices as geometric Brownian motion and investors problem as a stochastic differential equation. Merton's dynamic programming techniques along with Itô calculus derives explicit solutions for optimal and portfolio strategies under different utility functions, especially considering '*Hyperbolic Absolute Risk Aversion*'(HARA), and '*Constant Relative Risk Aversion*'(CARA). An important result by Merton called the 'Mutual Fund theorem' plays a prominent role in generalizing Markowitz's ideas in continuous-time setting, by separating optimal investment decisions into risk-free and risky investments [2] [8].

Taking the cue from Merton's research, this paper is organized as follows. Section 2 presents the model setup, introducing the financial market framework, including the risk-free and risky assets and the investor's wealth dynamics. Section 3 develops the analytical results, beginning with the derivation of the Hamilton–Jacobi–Bellman (HJB) equation, followed by the explicit solution for the optimal investment strategy and a comparative statics analysis of the optimal risky allocation. Section 4 focuses on computational results, where Monte Carlo simulations are used to generate asset price paths, analyze wealth dynamics under the optimal strategy, and compare performance with alternative strategies. Along with this section 5 and 6 gives a sensitivity analysis and empirical validation where feasible respectively. Finally, Section 7 summarizes the findings, discusses implications, and outlines directions for future research.

2 Model Formulation

We consider a continuous-time market on a finite time range $[0, T]$ comprising two assets:

- (a) A risk-free asset B_t satisfying

$$dB_t = rB_t dt,$$

where $r > 0$ is the constant risk-free interest rate.

- (b) A risky asset S_t following the geometric Brownian motion:

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

where $\mu \in \mathbb{R}$ is the drift, $\sigma > 0$ is the volatility, and $\{W_t\}_{t \geq 0}$ is a standard Brownian motion.

Let X_t denote the investor's wealth at time t . The control π_t denotes the amount (in dollars) invested in the risky asset at time t . Hence:

- π_t amount invested in the risky asset,
- $X_t - \pi_t$ amount invested in the risk-free asset.

2.1 Wealth Dynamics

Suppose the investor consumes at rate $c_t \geq 0$, then the self-financing wealth process satisfies

$$dX_t = [(r + \pi_t(\mu - r))X_t - c_t] dt + \pi_t \sigma X_t dW_t.$$

We assume no intermediate consumption, that is, when $c_t \equiv 0$, the wealth dynamics reduce to

$$dX_t = (r + \pi_t(\mu - r))X_t dt + \pi_t \sigma X_t dW_t.$$

2.2 Utility and Value Functions

We assume the investor has constant relative risk aversion (CRRA) utility. For terminal wealth, the utility function is

$$U(x) = \frac{x^{1-\gamma}}{1-\gamma}, \quad \gamma > 0, \gamma \neq 1.$$

The investor chooses an admissible portfolio process $\pi = \{\pi_t\}_{0 \leq t \leq T}$ to maximize expected utility of terminal wealth:

$$\sup_{\pi} \mathbb{E}[U(X_T)].$$

Thus, the value function is

$$V(t, x) = \sup_{\pi} \mathbb{E}_{t,x}[U(X_T)],$$

where $\mathbb{E}_{t,x}$ denotes conditional expectation given $X_t = x$.

3 Analytical Results

We derive the HJB equation and the explicit solution for the optimal portfolio strategy following the classical model introduced by Merton [2], with stochastic calculus tools as presented in standard texts such as [9, 10].

3.1 Derivation of the HJB equation

The HJB equation helps in characterizing the value function by reducing the dynamic optimization problem to a local maximization problem under certain assumptions. Merton's principle says that if the investor is currently at time t with wealth x , then the optimal value today should equal the best expected continuation value over a very short time interval $[t, t + \Delta t]$. Thus,

$$V(t, x) = \sup_{\pi} \mathbb{E}_{t,x}[V(t + \Delta t, X_{t+\Delta t})].$$

The introduction of the small interval $[t, t + \Delta t]$ allows the optimization problem to be analyzed locally in time. Applying Itô's Lemma to the infinitesimal change in the value function and letting $\Delta t \rightarrow 0$ leads to the Hamilton–Jacobi–Bellman equation. Note that we choose to optimize first over a small interval and then continue optimally afterward, instead of optimizing all the way to time T at once.

Itô's Lemma: Let $X(t)$ be an Itô process given by

$$dX(t) = \mu(t) dt + \sigma(t) dW(t).$$

Define

$$Y(t) = g(t, X(t)),$$

where the function g is differentiable in the first variable t , and twice-differentiable in the second variable. Then,

$$dY(t) = \frac{\partial g(t, X(t))}{\partial t} dt + \frac{\partial g(t, X(t))}{\partial x} dX(t) + \frac{1}{2} \frac{\partial^2 g(t, X(t))}{\partial x^2} (dX(t))^2.$$

Applying Itô's Lemma to V which depends on both time t and wealth X_t , we get,

$$dV(t, X_t) = V_t(t, X_t) dt + V_x(t, X_t) dX_t + \frac{1}{2} V_{xx}(t, X_t) (dX_t)^2.$$

We substitute the wealth dynamics

$$dX_t = (r + \pi_t(\mu - r)) X_t dt + \pi_t \sigma X_t dW_t.$$

Now compute $(dX_t)^2$. Since higher order dt terms are equal to 0 and $(dW_t)^2 = dt$, we get

$$(dX_t)^2 = (\pi_t \sigma X_t dW_t)^2 = \pi_t^2 \sigma^2 X_t^2 dt.$$

we now obtain

$$dV = \left[V_t + (r + \pi_t(\mu - r))X_t V_x + \frac{1}{2}\sigma^2 \pi_t^2 X_t^2 V_{xx} \right] dt + \pi_t \sigma X_t V_x dW_t.$$

Thus the infinitesimal change in the value function consists of:

- the drift term

$$\left[V_t + (r + \pi_t(\mu - r))X_t V_x + \frac{1}{2}\sigma^2 \pi_t^2 X_t^2 V_{xx} \right]$$

- the martingale term

$$\pi_t \sigma X_t V_x dW_t.$$

Now take conditional expectation given $X_t = x$. Since $\mathbb{E}[dW_t \mid X_t = x] = 0$, the martingale term disappears. Hence,

$$\mathbb{E}_{t,x}[dV(t, X_t)] = \left[V_t(t, x) + (r + \pi_t(\mu - r))x V_x(t, x) + \frac{1}{2}\sigma^2 \pi_t^2 x^2 V_{xx}(t, x) \right] dt.$$

In order to maximize the expected utility, we have to maximize the infinitesimal expected gain at every step. So, at each instant t we choose a suitable π_t to optimize the instantaneous expectation. Therefore,

$$0 = \sup_{\pi \in \mathbb{R}} \left\{ V_t(t, x) + (r + \pi(\mu - r))x V_x(t, x) + \frac{1}{2}\sigma^2 \pi^2 x^2 V_{xx}(t, x) \right\}.$$

Equivalently,

$$V_t(t, x) + \sup_{\pi \in \mathbb{R}} \left[(r + \pi(\mu - r))x V_x(t, x) + \frac{1}{2}\sigma^2 \pi^2 x^2 V_{xx}(t, x) \right] = 0.$$

This is the Hamilton–Jacobi–Bellman equation. Further, at time T there is no further control to choose, because the investor simply receives utility from terminal wealth. Hence,

$$V(T, x) = U(x) = \frac{x^{1-\gamma}}{1-\gamma}.$$

So the HJB equation is now,

$$V_t(t, x) + \sup_{\pi \in \mathbb{R}} \left[(r + \pi(\mu - r))x V_x(t, x) + \frac{1}{2}\sigma^2 \pi^2 x^2 V_{xx}(t, x) \right] = 0,$$

with terminal condition

$$V(T, x) = \frac{x^{1-\gamma}}{1-\gamma}.$$

3.2 Computation of the Optimizer

We now compute the optimizer inside the HJB. Define

$$F(\pi) = \pi(\mu - r)x V_x + \frac{1}{2}\sigma^2 \pi^2 x^2 V_{xx}.$$

Differentiate with respect to π :

$$\frac{\partial F}{\partial \pi} = (\mu - r)x V_x + \sigma^2 \pi x^2 V_{xx}.$$

Setting equal to zero:

$$(\mu - r)xV_x + \sigma^2 \pi^* x^2 V_{xx} = 0.$$

Thus,

$$\pi^* = -\frac{(\mu - r)V_x}{\sigma^2 x V_{xx}}.$$

To check this is indeed a maximum, compute

$$\frac{\partial^2 F}{\partial \pi^2} = \sigma^2 V_{xx}.$$

Under CRRA utility, the value function is concave in wealth, so typically

$$V_{xx} < 0.$$

Since $\sigma^2 > 0$, it follows that

$$\frac{\partial^2 F}{\partial \pi^2} < 0.$$

Therefore F is concave in π , and the critical point π^* gives the maximizing control.

3.3 Reduced HJB Equation

Substitute

$$\pi^* = -\frac{(\mu - r)V_x}{\sigma^2 x V_{xx}}$$

into the HJB. We compute the two π -dependent terms separately.

First,

$$\pi^*(\mu - r)xV_x = \left(-\frac{(\mu - r)V_x}{\sigma^2 x V_{xx}}\right)(\mu - r)xV_x = -\frac{(\mu - r)^2 V_x^2}{\sigma^2 V_{xx}}.$$

Second,

$$\frac{1}{2}\sigma^2(\pi^*)^2 x^2 V_{xx} = \frac{1}{2}\sigma^2 \left(\frac{(\mu - r)^2 x^2 V_x^2}{\sigma^4 x^2 V_{xx}^2}\right) x^2 V_{xx} = \frac{1}{2} \frac{(\mu - r)^2 V_x^2}{\sigma^2 V_{xx}}.$$

Adding these gives

$$\pi^*(\mu - r)xV_x + \frac{1}{2}\sigma^2(\pi^*)^2 x^2 V_{xx} = -\frac{(\mu - r)^2 V_x^2}{\sigma^2 V_{xx}} + \frac{1}{2} \frac{(\mu - r)^2 V_x^2}{\sigma^2 V_{xx}} = -\frac{1}{2} \frac{(\mu - r)^2 V_x^2}{\sigma^2 V_{xx}}.$$

Therefore the HJB reduces to

$$V_t + rxV_x - \frac{(\mu - r)^2}{2\sigma^2} \frac{V_x^2}{V_{xx}} = 0, \quad V(T, x) = \frac{x^{1-\gamma}}{1-\gamma}.$$

3.4 Homothetic Form of the Value Function

We now try to answer why the value function should have the form,

$$V(t, x) = \frac{A(t)}{1-\gamma} x^{1-\gamma}.$$

Firstly it is because the CRRA utility function is homogeneous of degree $1 - \gamma$:

$$U(\lambda x) = \frac{(\lambda x)^{1-\gamma}}{1-\gamma} = \lambda^{1-\gamma} U(x) \quad \text{for all } \lambda > 0.$$

Second, the wealth equation is linear in wealth. If the initial wealth is scaled by $\lambda > 0$ and the control is also scaled by λ , then the resulting wealth process is also scaled by λ .

Therefore the optimization problem inherits the same scaling:

$$V(t, \lambda x) = \lambda^{1-\gamma} V(t, x).$$

A function with this scaling property must have the form $V(t, x) = f(t) x^{1-\gamma}$ for some deterministic function $f(t)$. It is then convenient to write,

$$V(t, x) = \frac{A(t)}{1-\gamma} x^{1-\gamma}.$$

The terminal condition is given by

$$V(T, x) = U(x) = \frac{x^{1-\gamma}}{1-\gamma}.$$

Substituting $t = T$ into the ansatz, we obtain

$$V(T, x) = \frac{A(T)}{1-\gamma} x^{1-\gamma}.$$

Comparing the two expressions for $V(T, x)$, we have

$$\frac{A(T)}{1-\gamma} x^{1-\gamma} = \frac{x^{1-\gamma}}{1-\gamma}.$$

Simplifying further we obtain,

$$A(T) = 1.$$

3.5 Computation of Explicit Solution

Using the ansatz

$$V(t, x) = \frac{A(t)}{1-\gamma} x^{1-\gamma},$$

we compute:

$$V_t(t, x) = \frac{A'(t)}{1-\gamma} x^{1-\gamma},$$

$$V_x(t, x) = A(t) x^{-\gamma},$$

$$V_{xx}(t, x) = -\gamma A(t) x^{-\gamma-1}.$$

Now substitute these into the formula for π^* :

$$\pi^*(t, x) = -\frac{(\mu - r)V_x(t, x)}{\sigma^2 V_{xx}(t, x)} = -\frac{(\mu - r)A(t)x^{-\gamma}}{\sigma^2(-\gamma A(t)x^{-\gamma-1})}.$$

The factors $A(t)$ cancel, and

$$\frac{x^{-\gamma}}{x^{-\gamma-1}} = x.$$

Hence

$$\pi^*(t, x) = \frac{\mu - r}{\gamma \sigma^2} x.$$

Since the state variable is wealth, we write

$$\pi_t^* = \frac{\mu - r}{\gamma \sigma^2} X_t.$$

So the optimal amount invested in the risky asset is proportional to wealth. Dividing by wealth gives the optimal risky fraction:

$$\frac{\pi_t^*}{X_t} = \frac{\mu - r}{\gamma \sigma^2}.$$

Thus arises two formulas:

- if π_t means *amount (in dollars)*, then

$$\pi_t^* = \frac{\mu - r}{\gamma \sigma^2} X_t;$$

- if π_t means *fraction of wealth*, then

$$\pi_t^* = \frac{\mu - r}{\gamma \sigma^2}.$$

These formulas mean the same control written using two different notations.

3.6 Analytical Comparative Statics

The optimal risky fraction is

$$\frac{\pi_t^*}{X_t} = \frac{\mu - r}{\gamma \sigma^2}.$$

We now analyze analytically its dependency on the parameters.

Effect of risk aversion γ

$$\frac{\partial}{\partial \gamma} \left(\frac{\pi_t^*}{X_t} \right) = -\frac{\mu - r}{\gamma^2 \sigma^2} < 0 \quad \text{if } \mu > r.$$

Thus more risk aversion reduces the optimal proportion invested in the risky asset.

Effect of volatility σ

$$\frac{\partial}{\partial \sigma} \left(\frac{\pi_t^*}{X_t} \right) = -\frac{2(\mu - r)}{\gamma \sigma^3} < 0 \quad \text{if } \mu > r.$$

Thus higher volatility reduces the optimal risky allocation.

Effect of the risk premium $\mu - r$

$$\frac{\partial}{\partial (\mu - r)} \left(\frac{\pi_t^*}{X_t} \right) = \frac{1}{\gamma \sigma^2} > 0.$$

Thus a larger risk premium increases the optimal risky fraction.

Effect of time

The formula

$$\frac{\pi_t^*}{X_t} = \frac{\mu - r}{\gamma \sigma^2}$$

does not depend on t , so in the classical Merton model the optimal risky proportion is constant over time.

4 Data Analysis

4.1 Data Collection

We use the S&P 500 index, represented by the SPY ETF, as the risky asset in our simulation framework. The S&P 500 is a widely followed and highly liquid market benchmark containing large-cap U.S. equities. Due to its broad market representation and extensive historical data availability, it provides a suitable proxy for modeling risky asset dynamics in the Merton portfolio optimization problem.

Historical daily closing prices were collected using Microsoft Excel's built-in stock history functionality:

`=STOCKHISTORY("SPY", DATE(2026,1,1), DATE(2026,4,14))`

The selected sample period is:

January 1, 2026 – April 14, 2026.

The risk-free interest rate is approximated using the 3-month U.S. Treasury bill rate (DTB3), obtained from the Federal Reserve Economic Data (FRED) database. Treasury bills are commonly used as proxies for risk-free assets in continuous-time finance models.

The DTB3 data was obtained from:

<https://fred.stlouisfed.org/series/DTB3>

We compute a constant annualized risk-free rate by averaging the DTB3 observations over the same sample period used for the SPY data:

January 1, 2026 – April 14, 2026.

4.2 Data Cleaning

Historical stock prices: S&P: Based on the function in excel, the Close Price was only pulled for the dates between the specified date selection. Any day that the market was closed (ex. holidays) has an empty Close Price for that day. These days will not be considered and therefore, have been removed.

3-mo T-bill Rate Calculation: Remove any dates not within specified date selection. Any day that the market was closed has an empty DTB3 rate for that day. These rates will not be considered and therefore, have been removed.

4.3 Optimal Strategy Analysis

Using historical daily closing prices of the S&P 500 (SPY ETF), we compute log returns:

$$R_t = \ln \left(\frac{S_t}{S_{t-1}} \right).$$

The parameters are annualized as:

$$\mu = \text{avg}(R_t) \cdot 252, \quad \sigma = \text{std}(R_t) \cdot \sqrt{252}.$$

The risk-free rate is taken as the average 3-month Treasury bill rate:

$$r = \text{avg}(DTB3).$$

From the data, we obtain:

- Annualized Drift: $\mu = \text{avg}(R_t) \cdot 252 = 0.05986217869 \approx 5.99\%$
- Annualized Volatility: $\sigma = \text{std}(R_t) \cdot \sqrt{252} = 0.1492596969 \approx 14.93\%$
- Risk Free Rate: $r_f = \text{avg}(DTB3_t) = 0.03596760563 \approx 3.60\%$

To illustrate the model numerically, we consider the special case corresponding to logarithmic utility, obtained as the limiting case of the CRRA utility function as $\gamma \rightarrow 1$:

$$U(x) = \lim_{\gamma \rightarrow 1} \frac{x^{1-\gamma} - 1}{1-\gamma} = \ln x.$$

In this case, the optimal risky allocation simplifies to

$$\pi^* = \frac{\mu - r_f}{\sigma^2}.$$

Using the estimated market parameters, we obtain

$$\pi^* = \frac{0.0599 - 0.0360}{0.1493^2} \approx 1.0725.$$

Therefore, under the Merton Model, the optimal allocation is 1.0725, means that an investor should allocate 107.25% of their leveraged wealth to the risky asset, and -7.25% to the risk free asset. The negative weight signals to borrow 7.25% of wealth at the risk free rate.

4.4 Geometric Brownian Motion Simulation

The risky asset is modeled by the stochastic differential equation

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

where μ is the drift, $\sigma > 0$ is the volatility, and W_t is a standard Brownian motion.

To derive the discrete-time simulation formula, we apply Itô's Lemma to the function

$$g(s) = \ln s.$$

Let

$$Y_t = g(S_t) = \ln S_t.$$

Then

$$g'(s) = \frac{1}{s}, \quad g''(s) = -\frac{1}{s^2}.$$

By Itô's Lemma,

$$d(\ln S_t) = g'(S_t) dS_t + \frac{1}{2} g''(S_t) (dS_t)^2.$$

Since

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

we have

$$(dS_t)^2 = \sigma^2 S_t^2 dt.$$

Therefore,

$$\begin{aligned} d(\ln S_t) &= \frac{1}{S_t} (\mu S_t dt + \sigma S_t dW_t) + \frac{1}{2} \left(-\frac{1}{S_t^2} \right) \sigma^2 S_t^2 dt \\ &= \mu dt + \sigma dW_t - \frac{1}{2} \sigma^2 dt \\ &= \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dW_t. \end{aligned}$$

Integrating from t to $t + \Delta t$, we obtain

$$\ln S_{t+\Delta t} - \ln S_t = \left(\mu - \frac{1}{2} \sigma^2 \right) \Delta t + \sigma (W_{t+\Delta t} - W_t).$$

Since Brownian increments satisfy

$$W_{t+\Delta t} - W_t \sim N(0, \Delta t),$$

we may write

$$W_{t+\Delta t} - W_t = \sqrt{\Delta t} Z_t, \quad Z_t \sim N(0, 1).$$

Thus,

$$\ln S_{t+\Delta t} = \ln S_t + \left(\mu - \frac{1}{2} \sigma^2 \right) \Delta t + \sigma \sqrt{\Delta t} Z_t.$$

Exponentiating both sides gives the discrete-time simulation formula

$$S_{t+\Delta t} = S_t \exp \left[\left(\mu - \frac{1}{2} \sigma^2 \right) \Delta t + \sigma \sqrt{\Delta t} Z_t \right], \quad Z_t \sim N(0, 1).$$

4.5 Wealth Dynamics under the Optimal Strategy

Under the Merton framework, let X_t denote the investor's wealth process and let π denote the constant fraction of wealth invested in the risky asset. The wealth dynamics satisfy the stochastic differential equation

$$dX_t = X_t [\pi \mu + (1 - \pi) r] dt + X_t (\pi \sigma) dW_t.$$

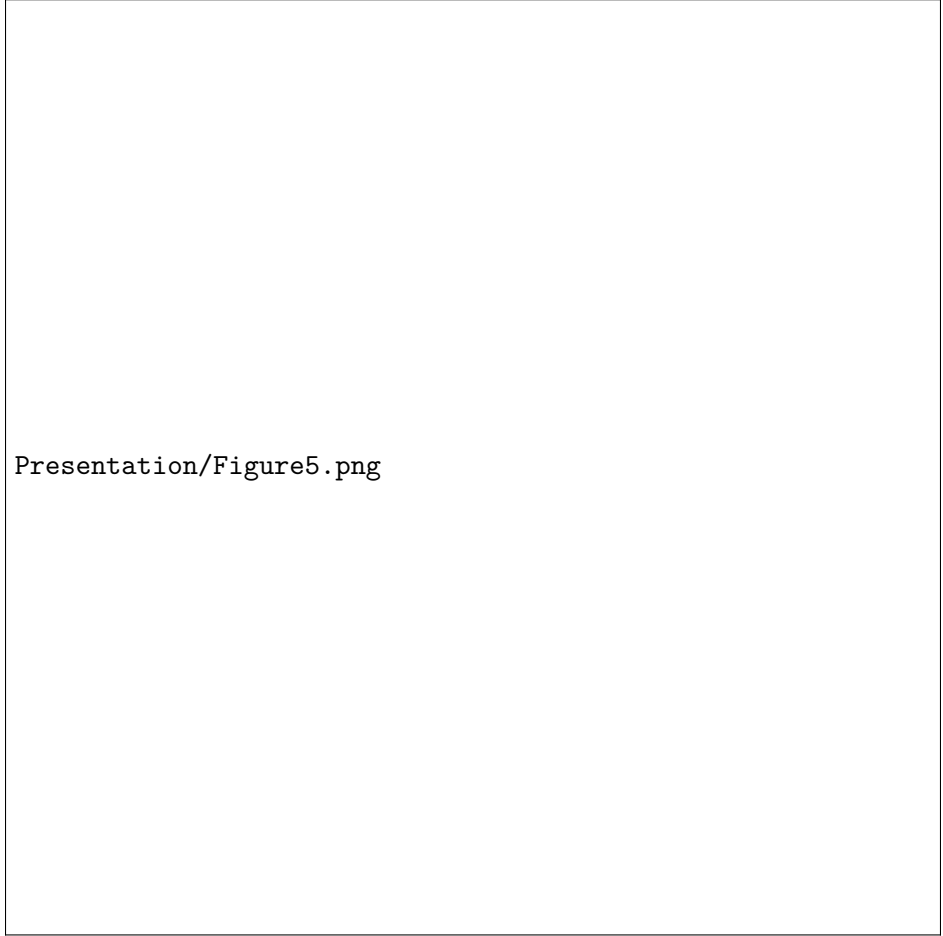
Using a similar process applied to the standard GBM in the above section with these modified parameters, we arrive at the discrete time approximation used in our simulation:

$$X_{t+\Delta t} = X_t \exp \left((\pi \mu + (1 - \pi) r - \frac{1}{2} \pi^2 \sigma^2) \Delta t + \pi \sigma \sqrt{\Delta t} Z_t \right).$$

The above SDE is discretized to get the wealth update equation:

$$W_{t+\Delta t} = W_t(1 + \pi R_t + (1 - \pi)r\Delta t)$$

The final row of the wealth matrix contains the terminal wealth for each simulated path. Plotting these values produces the terminal wealth distribution shown below.



Presentation/Figure5.png

Figure 1: Histogram showing the distribution of terminal wealth from all simulated paths.

4.6 Wealth Allocation with π^* compared to Other Allocation Strategies:

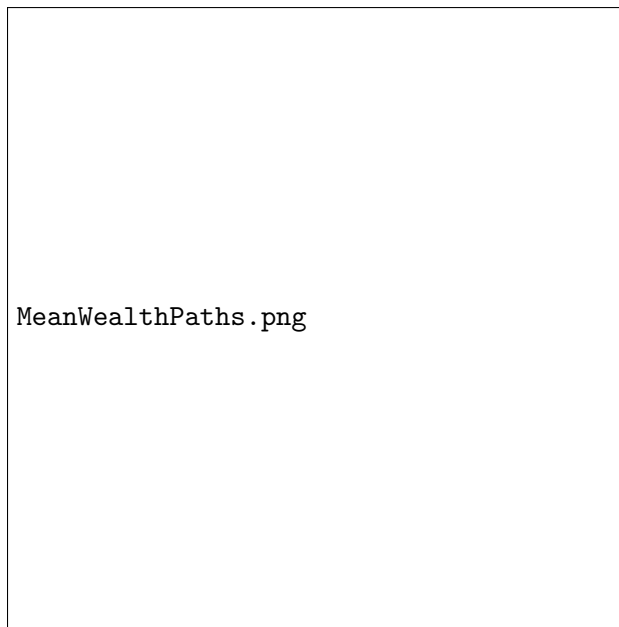


Figure 2: Comparison of mean wealth evolution under three investment strategies. The optimal strategy exhibits the fastest growth, the fixed 60% strategy shows moderate growth, and the cash strategy remains nearly flat.

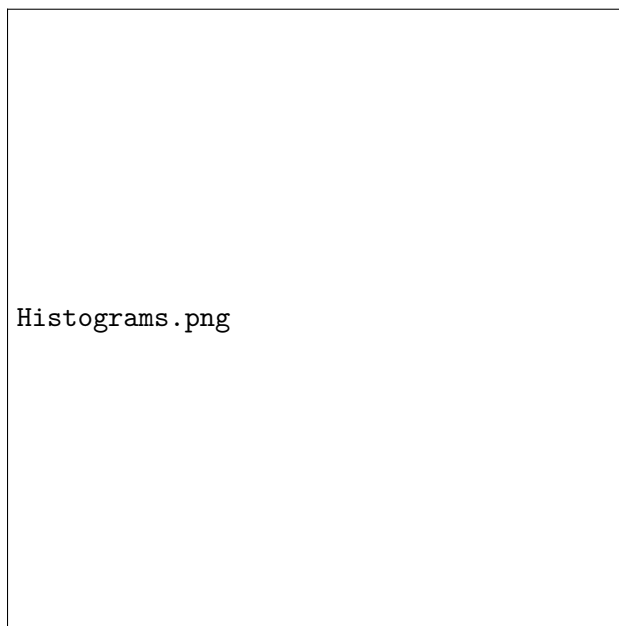


Figure 3: Terminal wealth distributions under the three strategies. The optimal strategy yields the highest average terminal wealth but also the widest spread, indicating greater risk. The fixed 60% strategy demonstrates moderate growth and lower volatility, while the cash strategy is nearly deterministic.

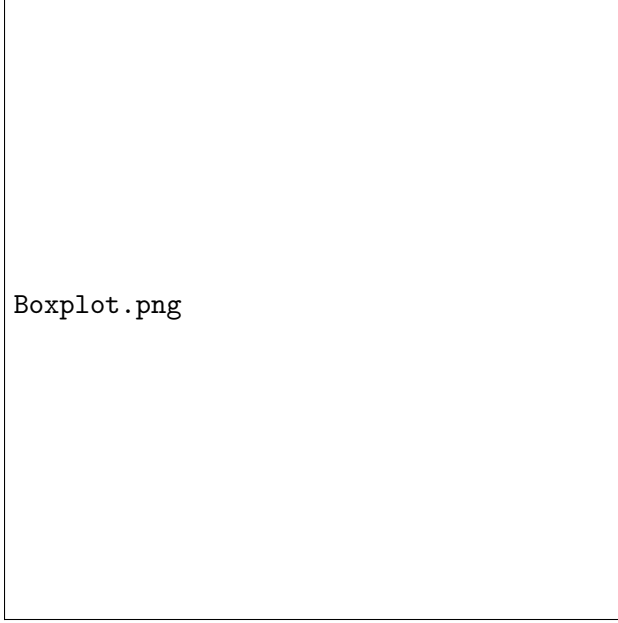


Figure 4: Comparison of terminal wealth variability across strategies. Higher-risk allocations exhibit wider dispersion, while the cash strategy remains stable with lower returns.

5 Welfare and Certainty Equivalent Sensitivity Analysis

5.1 Motivation

The sensitivity analysis above assumed that the model parameters are known when looking at the optimal portfolio allocation. In practice, however, we do not observe these parameters with certainty and need to estimate them from historical data. The estimation introduces errors, which may significantly affect investment decisions and the resulting welfare for an individual.

To quantify the effect of parameter misspecification, we study the welfare consequences of estimation error through the concept of certainty equivalents.

5.2 Certainty Equivalent Framework

A certainty equivalent represents the guaranteed amount of wealth that provides the same utility as a risky investment strategy. In other words, it converts uncertain future utility into an equivalent deterministic level of wealth today.

For a risk-averse investor, the certainty equivalent is generally lower than the expected terminal wealth, reflecting the investor's aversion to uncertainty. This framework provides a natural way to measure the welfare impact of incorrect parameter estimation, since it translates differences in expected utility into economically interpretable wealth losses.

5.3 Wealth Dynamics and Expected Growth

Assume that wealth X_t evolves according to the geometric Brownian motion

$$dX_t = X_t [r + \pi_t(\mu - r)] dt + X_t \pi_t \sigma dW_t.$$

Applying Itô's lemma to the logarithm of wealth gives

$$d(\ln X_t) = \left[r + \pi_t(\mu - r) - \frac{1}{2}\pi_t^2\sigma^2 \right] dt + \pi_t\sigma dW_t.$$

Integrating over the investment horizon $[0, T]$ and taking expectations yields

$$\mathbb{E}[\ln X_T] = \left[r + \pi_t(\mu - r) - \frac{1}{2}\pi_t^2\sigma^2 \right] T.$$

Define the growth rate function

$$g(\pi_t) = r + \pi_t(\mu - r) - \frac{1}{2}\gamma\pi_t^2\sigma^2.$$

The quadratic term arises from Itô's correction and represents the reduction in effective growth caused by volatility, often referred to as volatility drag.

For a CRRA investor, maximizing expected terminal utility is equivalent to maximizing this growth rate, so the value function takes the proportional form

$$V(\pi) \propto \exp(g(\pi)T).$$

5.4 Parameter Misspecification

Suppose the investor does not know the true expected return μ and instead uses the estimate

$$\hat{\mu} = \mu + \varepsilon_\mu.$$

The investor therefore chooses the portfolio weight

$$\hat{\pi} = \frac{\hat{\mu} - r}{\gamma\sigma^2} = \frac{(\mu - r) + \varepsilon_\mu}{\gamma\sigma^2} = \pi_t^* + \frac{\varepsilon_\mu}{\gamma\sigma^2},$$

where the optimal allocation under perfect information is

$$\pi_t^* = \frac{\mu - r}{\gamma\sigma^2}.$$

Define the deviation

$$\delta = \hat{\pi} - \pi_t^* = \frac{\varepsilon_\mu}{\gamma\sigma^2}.$$

5.5 Growth Rate Loss Under Misspecification

The reduction in growth rate caused by using $\hat{\pi}$ instead of the optimal allocation π_t^* is

$$\begin{aligned} g(\pi_t^*) - g(\hat{\pi}) &= \left[\pi_t^*(\mu - r) - \frac{1}{2}\gamma\pi_t^{*2}\sigma^2 \right] - \left[\hat{\pi}(\mu - r) - \frac{1}{2}\gamma\hat{\pi}^2\sigma^2 \right] \\ &= (\pi_t^* - \hat{\pi})(\mu - r) - \frac{1}{2}\gamma(\pi_t^{*2} - \hat{\pi}^2)\sigma^2. \end{aligned}$$

Substituting $\hat{\pi} = \pi_t^* + \delta$ gives

$$\pi_t^{*2} - \hat{\pi}^2 = -2\pi_t^*\delta - \delta^2.$$

Hence,

$$\begin{aligned} g(\pi_t^*) - g(\hat{\pi}) &= -\delta(\mu - r) + \frac{1}{2}\gamma(2\pi_t^*\delta + \delta^2)\sigma^2 \\ &= -\delta(\mu - r) + \gamma\pi_t^*\delta\sigma^2 + \frac{1}{2}\gamma\delta^2\sigma^2. \end{aligned}$$

Since

$$\pi_t^* = \frac{\mu - r}{\gamma\sigma^2},$$

we obtain

$$\gamma\pi_t^*\sigma^2 = \mu - r,$$

so the linear terms cancel, leaving

$$g(\pi_t^*) - g(\hat{\pi}) = \frac{1}{2}\gamma\sigma^2\delta^2.$$

Substituting the expression for δ gives

$$g(\pi_t^*) - g(\hat{\pi}) = \frac{1}{2}\gamma\sigma^2 \left(\frac{\varepsilon_\mu}{\gamma\sigma^2} \right)^2 = \frac{\varepsilon_\mu^2}{2\gamma\sigma^2}.$$

5.6 Welfare Loss via Certainty Equivalent

Over the horizon T , the corresponding value functions are

$$V(\pi_t^*) \propto \exp(g(\pi_t^*)T), \quad V(\hat{\pi}) \propto \exp(g(\hat{\pi})T).$$

For CRRA utility,

$$U(x) = \frac{x^{1-\gamma}}{1-\gamma},$$

the certainty equivalent wealth CE satisfies

$$U(CE) = V(\pi).$$

Thus,

$$\frac{CE^{1-\gamma}}{1-\gamma} = V(\pi).$$

Taking the ratio of certainty equivalents gives

$$\frac{CE(\hat{\pi})}{CE(\pi_t^*)} = \left(\frac{V(\hat{\pi})}{V(\pi_t^*)} \right)^{1/(1-\gamma)}.$$

Substituting the value function expressions yields

$$\frac{CE(\hat{\pi})}{CE(\pi_t^*)} = \exp \left(-\frac{\varepsilon_\mu^2}{2\gamma\sigma^2} \cdot \frac{T}{1-\gamma} \right).$$

Therefore, the fractional welfare loss is

$$\mathcal{L} = 1 - \frac{CE(\hat{\pi})}{CE(\pi_t^*)} = 1 - \exp\left(-\frac{\varepsilon_\mu^2}{2\gamma\sigma^2} \cdot \frac{T}{1-\gamma}\right).$$

Using the approximation

$$1 - e^{-x} \approx x \quad \text{for small } x,$$

we obtain

$$\mathcal{L} \approx \frac{\varepsilon_\mu^2}{2\gamma\sigma^2} \cdot \frac{T}{1-\gamma}.$$

5.7 Interpretation

The welfare loss expression provides several useful insights.

First, welfare loss grows quadratically with the estimation error ε_μ . This indicates that small estimation errors may be relatively harmless, but larger errors can become costly very quickly. Additionally, the dependence on the investment horizon T shows that the effects of incorrect parameter estimation accumulate over time. Longer investment periods amplify costs of wrongly estimating parameters. Finally, the presence of the risk aversion parameter γ in the denominator means that highly risk-averse investors are less sensitive to estimation errors, since they naturally invest less in risky assets.

Since variance, σ^2 , is in the denominator, wrongly estimating the variance is not as worrisome in this model when compared to getting μ wrong. A incorrect expected return is comparatively more harmful in low-volatility environments, where investors tend to take more risky positions.

6 Empirical Verification

6.1 Assets

- **Risky asset:** SPDR S&P 500 ETF (SPY) as a proxy for the market portfolio, with dividend-adjusted closing prices
- **Risk-free asset:** Fixed rate $r = 4\%$ per year, converted to a monthly equivalent

6.2 Sample Period and Frequency

- Sample period: February 1993 – December 2024
- Returns computed at monthly frequency via end-of-month resampling

6.3 Data Processing

- Daily adjusted closing prices downloaded using `yfinance` in Python
- Monthly returns computed as $r_t = \prod_{d \in t} (1 + r_d) - 1$ which just compounds daily returns to give a monthly result.

6.4 Rolling Parameter Estimation

- Using the Merton Rule derived in earlier sections, we estimate parameters each month using data from the year prior. Each prior month is weighted equally and averaged to find the new parameters.
- The risk-free rate is held fixed at $r = RFREE/12$ per month which gives a rate of 0.0033 per month.

6.5 Parameter Choices

Looking at the literature for other Merton solutions, a risk aversion of 3 is decently risk adverse, however, I chose to use a risk aversion of 0.5 to allow increased risk levels due to limiting ability to use leverage. The rolling window of a year was chosen to give enough data for the model to have an accurate estimate of market conditions, but not overweight past performance. Due to

- Risk aversion: $\gamma = 0.5$
- Rolling window: $W = 12$ months
- Re-balancing frequency: Monthly
- Leverage constraint: For this back-test, we assumed no leverage as individuals who are not institutional investors usually do not use leverage when investing.

6.6 Portfolio Return

At each month t , the portfolio return is:

$$r_t^p = w_t^* \cdot r_t^{\text{risky}} + (1 - w_t^*) \cdot r \quad (1)$$

6.7 Comparisons

To look at relative performance of the model, we chose to use the following strategies as comparison groups.

- **100% S&P 500:** Passive long-only equity exposure; represents the market portfolio
- **100% Bond:** Constant risk-free return $r/12$ each month; represents the no-risk baseline
- **50/50:** Equal-weighted portfolio rebalanced monthly; represents naive diversification with no optimization

6.8 Results

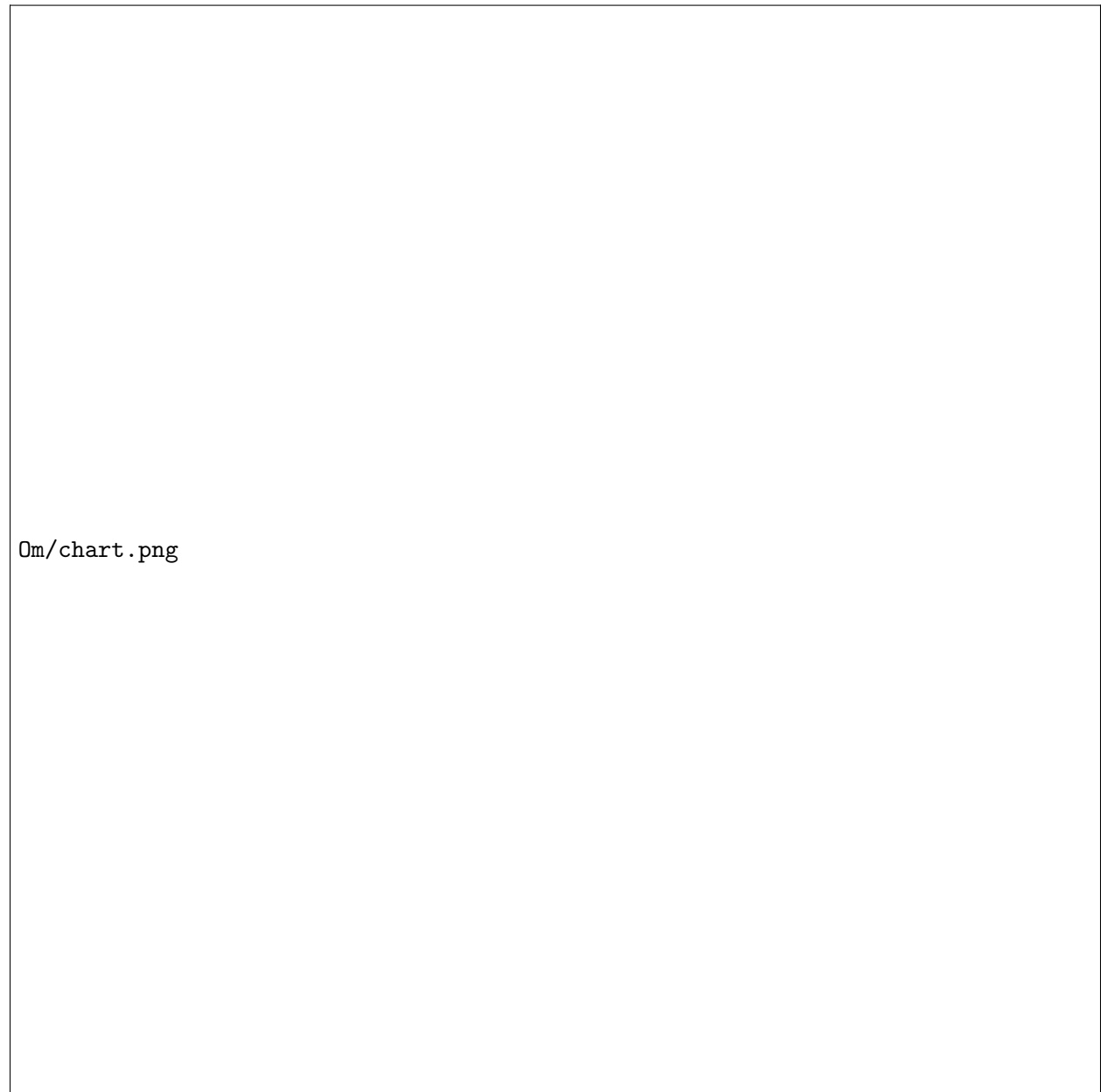


Figure 5: The following image shows how the rule did against competition historically.

6.9 Performance

The Merton model performed very well over time and beat the SPY with the preset parameters. Using different parameters, the results were similar as the model would perform in line with the SPY. When looking at the draw downs of the model, they typically only occur when there is large losses in the market such as the 2001 DotCOM bubble or COVID-19. Additionally, when compared to the SPY, the Merton model significantly reduces risk to downturns while competing with the SPY in total returns. When increasing the risk aversion, the Merton portfolio was still able to

compete with the SPY even with a lowered risk tolerance of 3.0.

6.10 Limitations

As discussed in the sensitivity analysis, this model is not perfect, especially in the real world for many reasons.

1. **Estimation error in μ :** Due to the noise in historical data, there would be a high chance that we mis-specified different parameters when estimating. This would lead to lowered performance of the model.
2. **Risk Free Rate:** For the sake of simplicity, we assumed a risk free rate of 4%, but this could be an off as the rate has varied from around 6% to 1%. The model assumes the ability to swap between the risk free rate and risky rate freely, but this is not always possible since the risk free rate changes over time.
3. **Bias in Risk Aversion:** Assuming a risk aversion of 3 is relatively arbitrary outside of academic standard and different parameters give vastly different results.
4. **Transaction Costs:** The model balances each month, however, no costs are associated with this. In reality, switching between risk free rates and risky rates would come with a cost and would lower total returns. This is not measured in the model.

7 Conclusion and Future work

This investigation of the classical Merton model to optimize portfolio established the following results:

- The Analytical derivation led to a constant allocation formula which depends fundamentally on market risk premium, asset volatility, and investor risk aversion.
- Data simulations confirmed our theoretical results i.e. the trade-off between higher expected returns and increased uncertainty in terminal wealth outcomes.
- Comparative strategies demonstrated that Merton's strategy under model assumptions is more advantageous than other strategies.
- Further, the welfare sensitivity analysis highlighted how estimation errors in expected returns can give us a meaningful certainty-equivalent losses, especially over longer investment time period.
- Even with certain drawbacks, the model provides realistic implications to empirical practice as increasing leverage for the Merton model would allow investors to increase their returns while maintaining a high sharp ratio to limit downside when compared to the S&P 500.

A natural extension of this work is to incorporate intermediate consumption, rate of interests, borrowing constraints etc. in order for the model to be more relevant to the real world market scenario. The assumption of constant drift and volatility could also be relaxed by considering stochastic volatility or regime-switching market models. Also, robust portfolio optimization can be studied in much detail. Finally, empirical validation can be conducted using longer historical datasets, diverse asset groups, and real portfolio back testing would provide stronger practical overview into

how well the classical Merton strategy performs in realistic financial markets. Additionally the use of multiple different risky assets and expanding the model to account for this addition would add more realism to the model. Including transaction costs into the model would also increase the applicability of the Merton rule.

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